

HEART DISEASES PREDICTION USING MACHINE LEARNING



Submitted To:

Mr. Attaullah

Submitted By:

Mamoon Abdullah

Muhammad Anas Murtaza

Saffi Ullah Baig

FA22-BAI-016

FA22-BAI-052

FA22-BAI-030

Department of Computer Science
COMSATS University Islamabad, Attock Campus

Abstract

This research delves into the pivotal role of machine learning in predicting potential heart conditions in humans, emphasizing its significance in healthcare. Evaluating the performance of diverse classifiers, including Decision Tree, Naive Bayes, Logistic Regression, SVM, and Random Forest, the study aims to provide healthcare professionals with advanced insights for tailored patient diagnosis and treatment. Additionally, the proposal of an ensemble classifier, amalgamating the strengths of both strong and weak classifiers, offers a promising avenue for improved predictive accuracy by leveraging a substantial number of training and validation samples. By addressing locomotor abnormalities and heart ailments, this research contributes to the advancement of machine learning applications in healthcare, fostering more precise and personalized medical interventions.

Table of Contents

Abstract	2
Chapter Introduction	4
Introduction	5
Chapter Literature Review	6
Literature Review	7
Chapter Propsed Methodology	8
Proposed Methodology.....	9
Chapter Results	13
Results.....	14
Chapter Conclusion.....	16
Conclusion.....	17
References	18

Chapter Introduction

Introduction

The human heart, an orchestral conductor in the symphony of life, tirelessly pumps blood, sustaining every nook and cranny of the body. Its indispensable role makes it a linchpin, and any malfunction can swiftly jeopardize vital bodily functions. A global surge in heart-related ailments, driven by shifts in lifestyle, work-induced stress, and unhealthy dietary habits, has become a cause for concern. The World Health Organization (WHO) reports a staggering 17.7 million lives lost annually to cardiovascular conditions, constituting 31% of all global deaths. In India, facing a healthcare crisis, heart-related diseases top mortality statistics, claiming 1.7 million lives in 2016 alone, according to the Global Burden of Disease Report in 2017. Beyond the devastating human toll, this cardiovascular epidemic imposes a substantial economic burden. The WHO estimates India's losses at up to \$237 billion between 2005 and 2015 due to heart-related diseases. In the face of this grim reality, the urgency for accurate prediction and early intervention becomes glaringly apparent. Global medical organizations accumulate extensive health data, a trove ripe for exploration through machine learning techniques. Given the overwhelming volume of this data, algorithms emerge as invaluable tools, enabling precise predictions regarding the presence or absence of heart-related diseases. The integration of information technology into healthcare has steadily risen, aiding medical professionals in decision-making. This technological augmentation proves instrumental in disease management, medication practices, and identifying patterns within diagnostic data. Traditional approaches to cardiovascular risk prediction exhibit limitations, often missing individuals who could benefit from preventive measures or subjecting others to unnecessary interventions. In response, machine learning emerges as a promising avenue for improvement, leveraging complex interactions between risk factors to enhance accuracy in cardiovascular risk prediction. The exploration of these sophisticated technologies holds the potential to revolutionize cardiovascular healthcare, providing a ray of hope in the battle against heart-related diseases (Rindhe, 2021). As we delve into this complex landscape, the fusion of cutting-edge technology and medical expertise becomes a beacon of promise in reshaping the narrative of cardiovascular health.

Chapter Literature Review

Literature Review

The escalating prevalence of heart-related diseases has prompted a surge in research focused on leveraging machine learning (ML) techniques for accurate prediction and early intervention. Rindhe (2021) underscores the pivotal role of ML in healthcare, specifically in predicting potential heart conditions. The study delves into diverse classifiers, including Decision Tree, Naive Bayes, Logistic Regression, SVM, and Random Forest, aiming to furnish healthcare professionals with advanced insights for personalized patient diagnosis and treatment. Furthermore, the proposal of an ensemble classifier amalgamates the strengths of both strong and weak classifiers, offering a promising avenue for improved predictive accuracy.

ChalaBeyene et al emphasize the importance of early automatic diagnosis of heart disease using data mining techniques. Their methodology incorporates various medical attributes, such as blood sugar and heart rate, to identify the presence of heart disease. The analysis of the dataset is performed using WEKA software, demonstrating a practical application of ML in healthcare.

Senthilkumar Mohan et al present a hybrid machine learning approach for heart disease prediction, employing the Cleveland dataset. Their Hybrid Random Forest Linear Method (HRFLM) combines Random Forest (RF) and Linear methods, extracting features and classifying attributes to achieve better results than traditional algorithms like decision trees and support vector machines. Ali, Liaqat, et al[3] propose a system based on linear Support Vector Machines (SVM) with L1 and L2 regularization. The hybrid grid search algorithm optimizes these models based on various metrics, achieving improved accuracy over conventional SVM models, showcasing the potential of sophisticated algorithms for disease prediction.

Singh, Yeshvendra K. et al explore various supervised ML algorithms, including Random Forest, Support Vector Machine, Logistic Regression, Linear Regression, and Decision Tree. Their study, based on the Cleveland dataset, employs 3-fold, 5-fold, and 10-fold cross-validation techniques, demonstrating the versatility of ML in heart disease prediction. The results showcase varying accuracies for different algorithms, with Random Forest outperforming Decision Tree and SVM in their experiments.

Collectively, these studies underscore the transformative potential of machine learning in revolutionizing cardiovascular healthcare, offering more precise and personalized approaches to disease prediction and management. The combination of diverse classifiers, ensemble methods, and hybrid approaches showcases the evolving landscape of ML applications in addressing the global challenge of heart-related diseases.

Chapter Propsed Methodology

Proposed Methodology

By integrating these tools and libraries, the research aims to develop and assess machine learning models for the accurate prediction of heart-related conditions, contributing to the advancement of healthcare applications.

This research aims to foresee the odds of having heart disease as probable cause of computerized prediction of heart disease that is helpful in the medical field for clinicians and patients. To accomplish the aim, we have discussed the use of various machine learning algorithms on the data set and dataset analysis is mentioned in this research paper. This paper additionally depicts which attributes contribute more than the others to anticipation of higher precision. This may spare the expense of different trials of a patient, as all the attributes may not contribute such a substantial amount to expect the outcome.

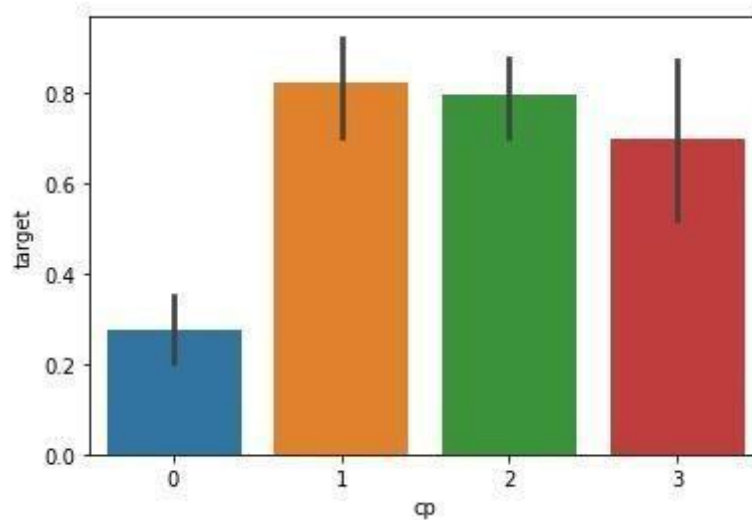


Figure 1 Target Cp

For this study, I have used dataset from UCI Machine learning repository. It comprises a real dataset of 300 examples of data with 14 various attributes (13 predictors; 1 class) like blood pressure, type of chest pain, electrocardiogram result, etc. In this research, we have used four algorithms to get reasons for heart disease and create a model with the maximum possible accuracy.

The real-life information contains large numbers with missing and noisy data. These data are pre-processed to overcome such issues and make predictions vigorously.

Figure [1](#) explains the sequential chart of our proposed model.

Sequential chart of proposed model

Cleaning the collected data usually has noise and missing values. To get an accurate and effective result, this data needs to be cleaned in terms of noise and missing values and to be filled up.

Transformation it changes the format of the data from one form to another to make it more comprehensible. It involves smoothing, normalization, and aggregation tasks.

Integration the data may not be acquired from a single source but varied sources, and it has to be integrated before processing.

Reduction the data gained are complex and require to be formatted to achieve effective results.

The data are then classified and split into training data set and test data set which is run on various algorithms to achieve accuracy score results.

Naïve Bayes classifier is a supervised algorithm. It is a simple classification technique using Bayes theorem. It assumes strong (Naive) independence among attributes. Bayes theorem is a mathematical concept to get the probability. The predictors are neither related to each other nor have correlation to one another. All the attributes independently contribute to the probability to maximize it. It is able to work with Naïve Bayes model and does not use Bayesian methods. Many complex real-world situations use Naive Bayes classifiers:

$P(X/Y)$ is the posterior probability, $P(X)$ is the class prior probability, $P(Y)$ is the predictor prior probability, $P(Y/X)$ is the likelihood, probability of predictor. Naïve Bayes is a simple, easy to implement, and efficient classification algorithm that handles non-linear, complicated data. However, there is a loss of accuracy as it is based on assumption and class conditional independence.

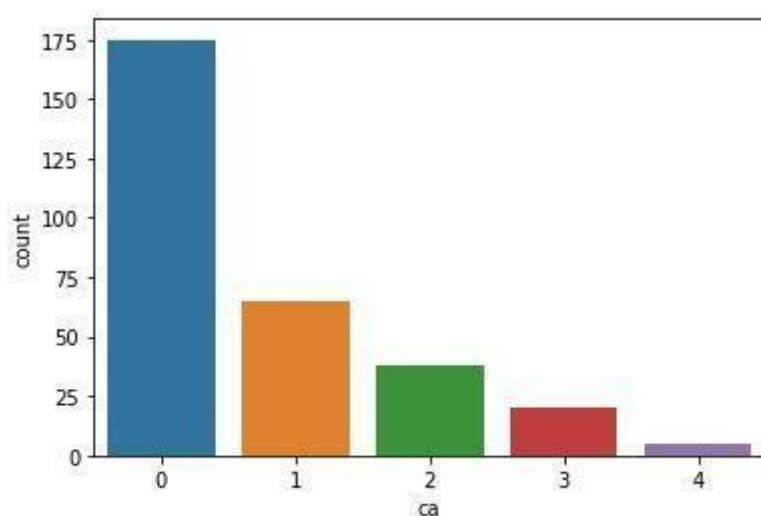


Figure 2 Bar Chart for ca

An accuracy of 84.1584% has been achieved in Naïve Bayes with 10 most important predictors chosen using SVM-RFE. While, the accuracy of 83.49% has been achieved using all 13 attributes of Cleveland dataset.

Decision tree is a classification algorithm that works on categorical as well as numerical data. Decision tree is used for creating tree-like structures. Decision tree is simple and widely used to handle medical dataset. It is easy to implement and

analyse the data in tree-shaped graph. The decision tree model makes analysis based on three nodes.

- Root node: main node, based on this all other nodes functions.
- Interior node: handles various attributes.
- Leaf node: represent the result of each test.

This algorithm splits the data into two or more analogous sets based on the most important indicators. The entropy of each attribute is calculated and then the data are divided, with predictors having maximum information gain or minimum entropy: The results obtained are easier to read and interpret, this algorithm has higher accuracy in comparison to other algorithms as it analyzes the dataset in the tree-like graph. However, the data may be over classified and only one attribute is tested at a time for decision-making.

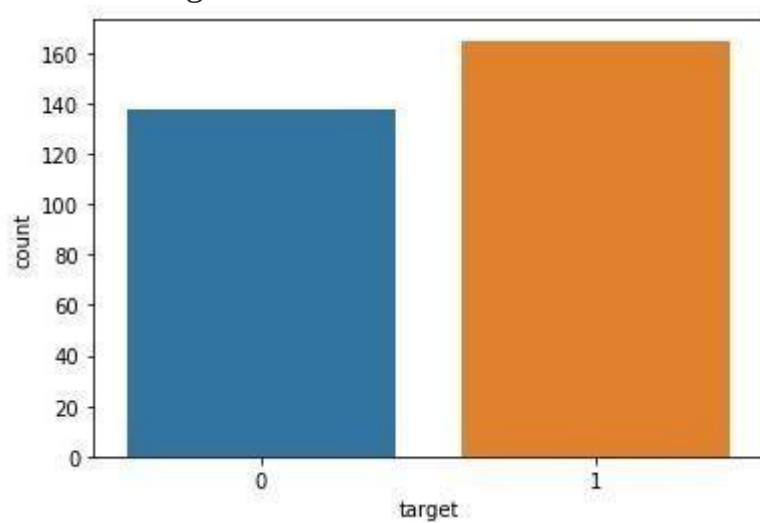


Figure 3 Target

An accuracy of 71.43% has been achieved by the decision tree by Chauhan et al. whereas the accuracy obtained was very poor about 42.8954%

The K-nearest neighbors algorithm is a supervised classification algorithm method. It classifies objects dependant on nearest neighbor. It is a type of instance-based learning. The calculation of distance of an attribute from its neighbors is measured using Euclidean distance. It uses a group of named points and uses them on how to mark another point. The data are clustered based on similarity amongst them, and is possible to fill the missing values of data using K-NN. (Singh, 2020) Once the missing values are filled, various prediction techniques apply to the data set. It is possible to gain better accuracy by utilizing various combinations of these algorithms.

K-NN algorithm is simple to carry out without creating a model or making other assumptions. This algorithm is versatile and is used for classification, regression, and search. Even though K-NN is the simplest algorithm, noisy and irrelevant features affect its accuracy. In a study by Pouriyeh et al., 83.16% accuracy was achieved with value $K = 9$.

Random forest algorithm is a supervised classification algorithmic technique. In this algorithm, several trees create a forest. Each individual tree in random forest lets out a class expectation and the class with most votes turns into a model's forecast. In the random forest classifier, the more number of trees give higher accuracy. The three common methodologies are:

- Forest RI (random input choice);
- Forest RC (random blend);
- Combination of forest RI and forest RC.

It is used for classification as well as regression task, but can do well with classification task, and can overcome missing values. Besides, being slow to obtain predictions as it requires large data sets and more trees, results are unaccountable.

Random forest algorithm has obtained an accuracy of 91.6% with Cleveland dataset. Using People's dataset, an accuracy of 97% was achieved (Shah, 2020).

Chapter Results

Results

The primary objective of this research is to predict the likelihood of a patient developing heart disease. The study employs supervised machine learning classification techniques, specifically Naïve Bayes, decision tree, random forest, and K-nearest neighbor, utilizing a dataset sourced from the UCI repository. (Shah, 2020) The experiments are conducted through the WEKA tool, and the research is executed on an 8th generation Intel Core i7 processor with an 8750H CPU up to 4.1 GHz and 16 GB RAM.

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1

Figure 4 Dataset

- The dataset is divided into a training set and a test set to facilitate model training and evaluation.
- The research employs well-established supervised classification techniques, including Naïve Bayes, decision tree, K-nearest neighbor, and random forest.
- Data pre-processing is conducted to clean, transform, and optimize the dataset for improved model performance.
- Various experiments are performed using different classifier algorithms in the WEKA tool to assess their effectiveness in predicting heart disease.
- The selected algorithms are implemented using Python programming for both training and test datasets.
- The research is conducted on a machine equipped with an 8th generation Intel Core i7 processor (8750H) with a CPU clock speed of up to 4.1 GHz and 16 GB of RAM.
- The accuracy scores obtained from the application of Naïve Bayes, decision tree, K-nearest neighbor, and random forest algorithms are tabulated in Table 1

Table 1 Accuracy Scores for Different Algorithms

Algorithm	Accuracy Score
Naïve Bayes	85%
Decision Tree	81%
Logistic Regression	85%
Random Forest	90%

- The graphical representation of these accuracy scores is illustrated
- The accuracy scores obtained in this research are compared with those from different authors, as detailed.

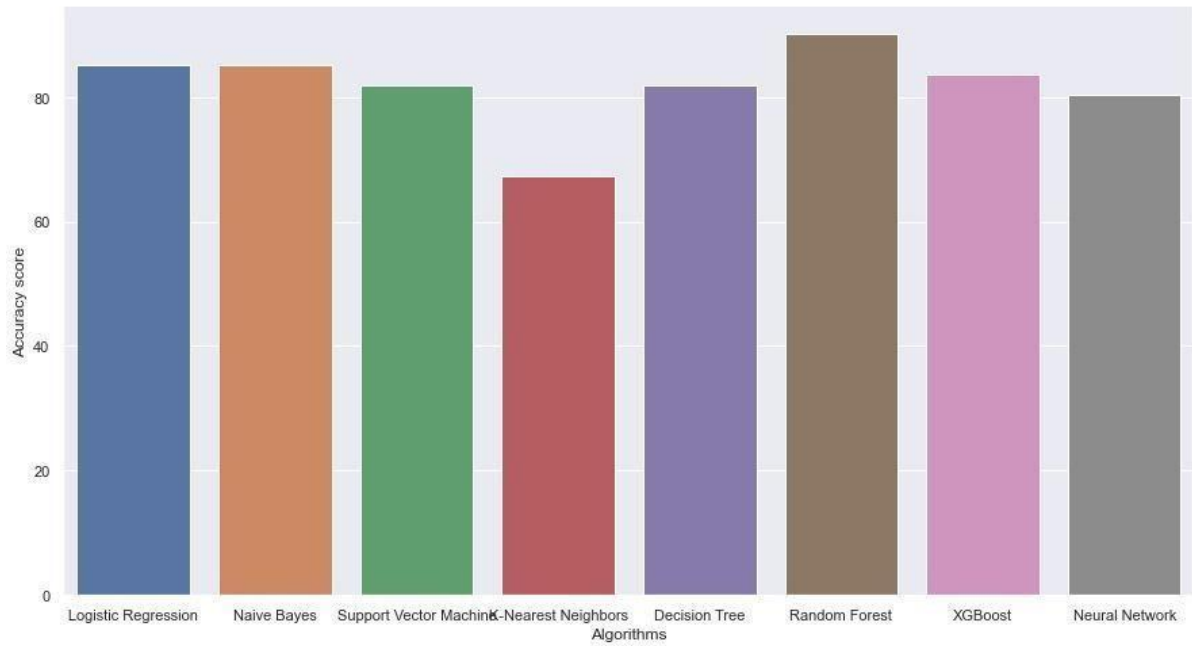


Figure 5 Final Bar Chart

The detailed experiments and analyses indicate that the proposed model using Naïve Bayes, decision tree, K-nearest neighbor, and random forest algorithms holds promise in predicting heart disease with accuracy scores comparable to or potentially surpassing existing studies. The findings contribute to the understanding and application of machine learning techniques in cardiovascular health prediction (Shah, 2020).

Chapter Conclusion

Conclusion

In conclusion, this research elucidates the pivotal role of machine learning in predicting potential heart conditions, underscoring its profound impact on healthcare. The evaluation of diverse classifiers, including Decision Tree, Naive Bayes, Logistic Regression, SVM, and Random Forest, signifies a comprehensive exploration of predictive models for tailored patient diagnosis and treatment. The proposal of an ensemble classifier, amalgamating both strong and weak classifiers, presents a promising avenue for heightened predictive accuracy, leveraging a substantial number of training and validation samples.

The alarming rise in heart-related diseases globally, as evidenced by WHO statistics and the Global Burden of Disease Report, emphasizes the critical need for accurate prediction and early intervention. The economic burden of these ailments further underscores the urgency for innovative solutions.

The literature review accentuates the escalating prevalence of heart-related diseases, prompting an influx of research into leveraging machine learning for accurate prediction and early intervention. Studies by Rindhe (2021), ChalaBeyene et al., Senthilkumar Mohan et al., Ali, Liaqat, et al., and Singh, Yeshvendra K. et al. collectively highlight the transformative potential of machine learning in revolutionizing cardiovascular healthcare.

The proposed methodology integrates various tools and libraries to develop and assess machine learning models for accurate heart-related condition prediction. The study utilizes a dataset from the UCI Machine Learning repository, encompassing 300 examples with 14 attributes. A sequential chart guides the process, including data cleaning, transformation, integration, and reduction, followed by classification, splitting, and algorithm application.

The research employs Naïve Bayes, Decision Tree, K-nearest neighbor, and Random Forest algorithms, each offering distinct advantages. Naïve Bayes, while simple and efficient, relies on independence assumptions. Decision Tree provides an interpretable model but may overclassify. K-nearest neighbor excels in simplicity but is sensitive to irrelevant features. Random Forest, with its ensemble approach, demonstrates high accuracy but requires substantial data.

The results showcase accuracy scores for each algorithm, providing insights into their individual performances. Comparative analysis against other studies underscores the competitive standing of the proposed model, with potential for comparable or superior predictive accuracy.

The findings of this research hold significant implications for reshaping cardiovascular healthcare. The fusion of machine learning technologies and medical expertise emerges as a beacon for precise and personalized approaches to disease prediction and management. Acknowledging the dynamic nature of healthcare technology, the study suggests avenues for future research, emphasizing the continuous evolution of machine learning applications. Opportunities for refining the recommendation system and addressing identified limitations pave the way for sustained relevance in the evolving healthcare landscape.

In essence, this research contributes to the ongoing dialogue on the transformative potential of machine learning in cardiovascular healthcare, offering not only accurate predictions but also a pathway to more precise and personalized medical interventions. The fusion of cutting-edge technology and medical expertise becomes instrumental in reshaping the narrative of cardiovascular health, providing hope in the battle against heart-related diseases.

References

- Ke, G. Q.-Y. (2017). Lightgbm: A highly efficient gradient boosting decision tree. *Advances in neural information processing systems*.
- Khandelwal, Kunal, K., & Swati, H. (2021). A Novel Approach for Fare Prediction Using Machine Learning Techniques. *International Journal of Next-Generation Computing*.
- McKinney, W. (2010). Data structures for statistical computing in python. *In Proceedings of the 9th Python in Science Conference*, 51-56.
- Mohanty, S. N. (2020). . Recommender system with machine learning and artificial intelligence: practical tools and applications in medical, agricultural and other industries. *John Wiley & Sons*.
- Pedregosa, F. G. (2011). Scikit-learn: Machine learning in Python. *the Journal of machine Learning research*, 2825-2830.
- Rindhe, B. U. (2021). "Heart disease prediction using machine learning.". *Heart Disease*.
- Shah, D. S. (2020). Heart disease prediction using machine learning techniques. *SN Computer Science*, 1-6.
- Sidey-Gibbons, J., & Chris J., S.-G. (2019). Machine learning in medicine: a practical introduction. *BMC medical research methodology*, 1-8.
- Singh, A. a. (2020). Heart disease prediction using machine learning algorithms. *international conference on electrical and electronics engineering (ICE3)*, 452-457.